On semi-relaxed optimal transport

Anonymous Author(s)
Affiliation
Address
email

Abstract

Semi-relaxed optimal transport generalizes the classical optimal transport problem by relaxing one of the marginal constraints. While standard approaches rely on linear programming, they suffer from significant computational complexity. In this paper, we propose a novel regularized formulation for semi-relaxed optimal transport that leads to an efficient Sinkhorn-like alternating maximization scheme. We provide a rigorous dual formulation in the log-domain and prove strong duality. Furthermore, we establish the existence and uniqueness of the dual potentials and demonstrate the linear convergence of both the potentials and the primal objective. Our theoretical guarantees do not require finite space assumptions, extending the applicability of these methods to more general settings.

1 Introduction

Optimal transport has emerged as a powerful tool in machine learning and applied mathematics for comparing probability distributions. However, in many practical scenarios, the exact matching of both marginals is too restrictive. Semi-relaxed optimal transport addresses this limitation by strictly enforcing only one marginal constraint while relaxing the other, often using a divergence penalty. This formulation is particularly relevant for applications where the target distribution is only partially known or subject to noise.

1.1 Background and Related Works

Notations and preliminaries. Given a function $f : \mathcal{X} \to \mathcal{Y}$, its effective domain is denoted by $\mathrm{dom}(f) \triangleq \{x \in \mathcal{X} : f(x) < +\infty\}$; and its range is denoted by $\mathrm{Ran}(f) \triangleq \{f(x) : x \in \mathcal{X}\}$. Given a set $\mathcal{A}$, its interior is denoted by $\mathrm{int}(\mathcal{A})$. Given two measures μ and ν, the notation $\mu \ll \nu$ signifies that μ is absolutely continuous with respect to ν. Across this paper, $\mathcal{X}$ and $\mathcal{Y}$ are Polish spaces. The set of probability measures over $\mathcal{X}$ equipped with the σ-field of Borel sets is denoted by $\triangle(\mathcal{X})$. The set of p-integrable functions with respect to P is denoted by $\mathrm{L}^p(\mathcal{X}, P)$. Given two probability measures $P \in \triangle(\mathcal{X})$ and $Q \in \triangle(\mathcal{Y})$, the notation $\Pi(P, Q)$ denotes the set of probability measures on $\mathcal{X} \times \mathcal{Y}$ with marginals P and Q in $\triangle(\mathcal{X})$ and $\triangle(\mathcal{Y})$, respectively. The probability measures in $\Pi(P, Q)$ are called couplings. By extension, $\Pi(P, \star)$ denotes the set of probability measures on $\mathcal{X} \times \mathcal{Y}$ where only the first marginal is constrained to be P. The product measure of $P \in \triangle(\mathcal{X})$ and $Q \in \triangle(\mathcal{Y})$ in $\triangle(\mathcal{X} \times \mathcal{Y})$ is denoted by PQ. The first marginal of a measure $\pi \in \triangle(\mathcal{X} \times \mathcal{Y})$ is denoted by $\pi^{\mathcal{X}}_{\#}$. Similarly, its second marginal is denoted by $\pi^{\mathcal{Y}}_{\#}$. Given two functions $f : \mathcal{X} \to \mathbb{R}$ and $g : \mathcal{X} \to \mathbb{R}$, the notation $f \overset{a.s.}{\underset{P}{=}} g$ means that f and g are almost surely equal with respect to P. Similarly, the notation $f \overset{a.s.}{\underset{P}{\geq}} g$ means that f is greater than g almost surely with respect to P. Given $\mathcal{A}$ and $\mathcal{B}$, two subsets of $\mathcal{X}$, the notation $\mathcal{A} \overset{a.s.}{\underset{P}{\subset}} \mathcal{B}$ means that $P(\mathcal{A} \setminus \mathcal{B}) = 0$. Let $\mathcal{E}$ be a Banach space and $\mathcal{E}'$ be its topological dual space. Given $x \in \mathcal{E}$ and $y \in \mathcal{E}'$, $\langle y, x \rangle$ denotes the dual pairing between

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

y and x. If $f : \mathcal{X} \to \mathbb{R}$ and $g : \mathcal{Y} \to \mathbb{R}$ are two real functions then $f \oplus g : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$ is the function defined through $f \oplus g(x, y) = f(x) + g(y)$. The following definition formalizes the notion of admissibility for an optimization problem, ensuring that candidates yield a finite value within the functional domain.

Definition 1.1. *Given a set $\mathcal{M}$ and a functional $R : \mathcal{M} \to \mathbb{R} \cup \{+\infty\}$, an instance $\theta \in \mathcal{M}$ is said to be admissible for the following problem*

$$\inf_{\nu \in \mathcal{M}} R(\nu), \tag{1}$$

if $R(\theta) < +\infty$.

Analysis of optimality conditions and the derivation of dual problems rely heavily on the transformation of functions into a dual space representation. This operation is performed through the convex conjugate, providing a bridge between primal and dual variables in Banach spaces. The convex conjugate is defined below.

Definition 1.2. *Let $\mathcal{E}$ be a Banach space and $\mathcal{E}'$ be its topological dual space. Let $\phi : \mathcal{E} \to \mathbb{R} \cup \{-\infty, +\infty\}$; the convex conjugate (or Fenchel conjugate) of ϕ is the function $\phi^* : \mathcal{E}' \to \mathbb{R} \cup \{-\infty, +\infty\}$ defined for $y \in \mathcal{E}'$ through (see [Zalinescu, 2002, Section 2.3])*

$$\phi^*(y) = \sup_{x \in \mathcal{E}} (\langle y, x \rangle - \phi(x)). \tag{2}$$

Complementary to the convex conjugate, the indicator function serves as a fundamental tool through which constraints are encoded directly into the objective functional. This formulation is particularly suited for optimal transport problems, as it ensures the strict enforcement of marginal constraints. The following definition formalises this concept, providing a mechanism to assign an infinite cost to any instance outside the feasible set.

Definition 1.3. *Let $\mathcal{C}$ be a subset of a set $\mathcal{E}$. The indicator function of $\mathcal{C}$, denoted through $\iota_{\mathcal{C}}$, is defined as:*

$$\iota_{\mathcal{C}}(x) = \begin{cases} 0 & \text{if } x \in \mathcal{C}; \\ +\infty & \text{if } x \notin \mathcal{C}. \end{cases} \tag{3}$$

2 Problem Formulation: Semi-Relaxed Optimal Transport

Throughout this section, $(\mathcal{X}, P)$ and $(\mathcal{Y}, Q)$ are fixed probability spaces. The function $\mathsf{c} : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$ is a lower semi-continuous measurable function. Consider the problem of semi-relaxed optimal transport, defined as follows

$$\inf_{\pi \in \Pi(P, \star)} \int_{\mathcal{X} \times \mathcal{Y}} \mathsf{c}(x, y) \mathrm{d}\pi(x, y). \tag{4}$$

Standard approaches for solving semi-constrained optimal transport problems often rely on linear programming Cuturi [2013]. However, these methods involve significant computational complexity, which frequently becomes prohibitive for large-scale datasets or high-dimensional problems. Such linear programming formulations scale poorly with the number of support points, making them computationally intractable for many modern machine learning applications.

3 Sinkhorn-Like Regularization Scheme

Throughout this section, $(\mathcal{X}, P)$ and $(\mathcal{Y}, Q)$ are fixed probability spaces. The function $\mathsf{c} : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$ is a lower semi-continuous, measurable, bounded function, and α and β are strictly positive numbers. The objective is to extend the results established in the seminal work of Cuturi [2013] to the framework of semi-relaxed optimal transport. This goal is achieved through a specific choice of regularization that ensures the dual problem is amenable to Sinkhorn-like iterations. The proposed scheme fits within the general framework introduced in Chizat et al. [2018]. More precisely, the problem is a particular instance of [Chizat et al., 2018, Equation 3.1] where $n = 1$, the functional $\mathcal{F}_1$ is the indicator function of the set $\{P\}$, and $\mathcal{F}_2$ equals β times the Kullback-Leibler (KL) divergence with respect to Q, with β strictly positive. The reference measures, denoted as dx and dy in Chizat et al. [2018], correspond here to P and Q, respectively. This specialization leads to the following problem

$$\inf_{\pi \in \Pi(P, \star)} \mathsf{P}(\pi) \triangleq \int_{\mathcal{X} \times \mathcal{Y}} \mathsf{c}(x, y) \mathrm{d}\pi(x, y) + \alpha D_{\mathrm{KL}}(\pi || PQ) + \beta D_{\mathrm{KL}}(\pi^{\mathcal{Y}}_{\#} || Q). \tag{5}$$

3.1 Connections to Existing Dual Formulations

From [Chizat et al., 2018, Theorem 3.2], the dual of problem (5) can be written as follows

$$\sup_{u\in \mathrm{L}^\infty(\mathcal{X}), v\in \mathrm{L}^\infty(\mathcal{Y})} \int_{\mathcal{X}} u(x)\mathrm{d}P(x) - \beta \int_{\mathcal{Y}} \left(e^{-\frac{v(y)}{\beta}} - 1\right) \mathrm{d}Q(y) \\ -\alpha \int_{\mathcal{X}\times\mathcal{Y}} \left(e^{\frac{u(x)+v(y)-\mathsf{c}(x,y)}{\alpha}} - e^{-\frac{\mathsf{c}(x,y)}{\alpha}}\right) \mathrm{d}PQ(x,y). \tag{6}$$

The theoretical results from Chizat et al. [2018] remain applicable to this dual formulation. In particular, strong duality holds, ensuring equality between the supremum in (6) and the infimum in (5). The convergence properties established in [Chizat et al., 2018, Proposition 3.9, Theorem 4.1] are also valid. Nevertheless, this work aims to provide enhanced convergence results and refined theoretical guarantees. From [Chizat et al., 2018, Section 3.3, Table 1], the iterates obtained by alternating maximization on D in (6) can be written as follows

$$u^{(l+1)} = -\alpha \log\left(\int_{\mathcal{Y}} e^{\frac{v^{(l)}(y)-\mathsf{c}(x,y)}{\alpha}} \mathrm{d}Q(y)\right); \tag{7a}$$

$$v^{(l+1)} = -\frac{\alpha\beta}{\alpha+\beta} \log\left(\int_{\mathcal{X}} e^{\frac{u^{(l+1)}(x)-\mathsf{c}(x,y)}{\alpha}} \mathrm{d}P(x)\right). \tag{7b}$$

Or alternatively by considering the so-called scaled iterations

$$(a^{(l)}, b^{(l)}) \triangleq (e^{\frac{u^{(l)}}{\alpha}}, e^{\frac{v^{(l)}}{\alpha}}), \tag{8}$$

for which the algorithm is written as follows

$$a^{(k+1)}(x) = \left(\int_{\mathcal{Y}} e^{-\frac{\mathsf{D}(x,y)}{\alpha}}\, b^{(k)}(y)^{-\frac{\beta}{\alpha}}\, \mathrm{d}Q(y)\right)^{-1}; \tag{9a}$$

$$b^{(k+1)}(y) = \left(\int_{\mathcal{X}} e^{-\frac{\mathsf{D}(x,y)}{\alpha}}\, a^{(k+1)}(x)\, \mathrm{d}P(x)\right)^{\frac{\alpha}{\alpha+\beta}}. \tag{9b}$$

While these results hold in a highly general framework, it is possible to achieve tighter theoretical bounds by explicitly exploiting the structure of our semi-relaxed problem, particularly the fact that the feasible couplings are probability measures.

3.2 A Novel Functional Dual Formulation

This section proposes an alternative dual formulation aiming to strengthen the convergence results of the potentials without requiring finite space assumptions. The derivation in Chizat et al. [2018] is highly general and does not explicitly account for the fact that the feasible couplings for the primal problems are necessarily probability measures. The alternative derivation presented here utilizes the variational formula of the KL divergence. The dual problem is subsequently obtained through the inversion of the infimum and supremum. This approach results in a formulation situated in the log-domain, which shares structural similarities with the notion of free energy in statistical physics Di Marino and Gerolin [2020].

$$\sup_{u\in \mathrm{L}^\infty(\mathcal{X}), v\in \mathrm{L}^\infty(\mathcal{Y})} \mathsf{D}(u,v) \triangleq \int_{\mathcal{X}} u(x)\mathrm{d}P(x) - \beta \log\left(\int_{\mathcal{Y}} e^{-\frac{v(y)}{\beta}} \mathrm{d}Q(y)\right) \\ -\alpha \log\left(\int_{\mathcal{X}\times\mathcal{Y}} e^{\frac{u(x)+v(y)-\mathsf{c}(x,y)}{\alpha}} \mathrm{d}PQ(x,y)\right). \tag{10}$$

3.2.1 Weak Duality

Lemma 3.1. *Weak duality holds between the primal* (5) *and the dual problem* (10)*, that is*

$$\sup_{u\in \mathrm{L}^\infty(\mathcal{X}), v\in \mathrm{L}^\infty(\mathcal{Y})} \mathsf{D}(u,v) \leq \inf_{\pi\in\Pi(P,\star)} \mathsf{P}(\pi) < +\infty. \tag{11}$$

The proof is detailed in Appendix A. This weak duality result implies in particular that the value of the dual problem is strictly finite.

3.2.2 Dual Attainment

This alternative dual formulation, while appearing different, yields an alternating maximization scheme closely resembling the standard one. The following property reflects the fact that the functional D is not strictly concave, meaning the maximizing potentials are not unique. However, this property is instrumental in proving their existence.

Lemma 3.2. *For all functions $u \in \mathrm{L}^\infty(\mathcal{X}), v \in \mathrm{L}^\infty(\mathcal{Y})$ and real numbers M, N, the dual functional D in* (10) *satisfies*

$$\mathsf{D}(u, v) = \mathsf{D}(u + M, v) = \mathsf{D}(u, v + N). \tag{12}$$

The proof is detailed in Appendix A.

Lemma 3.3. *The dual functional D in* (10) *is differentiable on $L^\infty(\mathcal{X}) \times L^\infty(\mathcal{Y})$. Moreover, for all $(u, v) \in L^\infty(\mathcal{X}) \times L^\infty(\mathcal{Y})$ and directions $(a, b) \in L^\infty(\mathcal{X}) \times L^\infty(\mathcal{Y})$, its partial derivatives are given by*

$$\langle \frac{\partial \mathsf{D}(u,v)}{\partial u}, a\rangle = \int_{\mathcal{X}} a(x) \left(1 - \frac{1}{Z(u,v)} \int_{\mathcal{Y}} e^{\frac{u(x)+v(y)-\mathsf{c}(x,y)}{\alpha}} \mathrm{d}Q(y)\right) \mathrm{d}P(x); \tag{13}$$

$$\langle \frac{\partial \mathsf{D}(u,v)}{\partial v}, b\rangle = \int_{\mathcal{Y}} b(y) \left(\frac{e^{-\frac{v(y)}{\beta}}}{\int_{\mathcal{Y}} e^{-\frac{v(y)}{\beta}} \mathrm{d}Q(y)} - \frac{1}{Z(u,v)} \int_{\mathcal{X}} e^{\frac{u(x)+v(y)-\mathsf{c}(x,y)}{\alpha}} \mathrm{d}P(x)\right) \mathrm{d}Q(y), \tag{14}$$

where $Z(u, v) \triangleq \int_{\mathcal{X}\times\mathcal{Y}} \exp\left(\frac{u(x)+v(y)-\mathsf{D}(x,y)}{\alpha}\right) \mathrm{d}PQ(x, y)$.

The proof is detailed in Appendix A.

By setting the partial derivatives to zero, and simultaneously enforcing a normalization condition, we obtain uniqueness of the updates and a simplified expression for the primal solution.

Corollary 3.1. *Let $(\hat{u}, \hat{v})$ be in $\mathrm{L}^\infty(\mathcal{X}) \times \mathrm{L}^\infty(\mathcal{Y})$. If for all $x \in \mathcal{X}$, u satisfies*

$$\hat{u}(x) = \Phi(\hat{v})(x); \tag{15}$$

$$\text{with } \Phi(\hat{v})(x) \triangleq -\alpha \log \left(\int_{\mathcal{Y}} e^{\frac{\hat{v}(y)-\mathsf{c}(x,y)}{\alpha}} \mathrm{d}Q(y)\right), \tag{16}$$

then, $\hat{u}$ maximizes $u \mapsto \mathsf{D}(u, \hat{v})$, where D is in (10)*. Similarly, if for all $y \in \mathcal{Y}$, v satisfies*

$$\hat{v}(y) = \Psi(\hat{u})(y) : \tag{17}$$

$$\text{with } \Psi(\hat{u})(y) \triangleq \beta \log \left(\int_{\mathcal{Y}} \left(\int_{\mathcal{X}} e^{\frac{\hat{u}(x)-\mathsf{c}(x,y')}{\alpha}} \mathrm{d}P(x)\right)^{\frac{\alpha}{\alpha+\beta}} \mathrm{d}Q(y')\right) - \frac{\alpha\beta}{\alpha+\beta} \log \left(\int_{\mathcal{X}} e^{\frac{\hat{u}(x)-\mathsf{c}(x,y)}{\alpha}} \mathrm{d}P(x)\right), \tag{18}$$

then, $\hat{v}$ maximizes $v \mapsto \mathsf{D}(\hat{u}, v)$. Moreover, if (15) *and* (17) *are simultaneously satisfied, then $(\hat{u}, \hat{v})$ maximizes $(u, v) \mapsto \mathsf{D}(u, v)$.*

The proof is detailed in Appendix A.

To arrive at the compactness arguments necessary for proving existence Di Marino and Gerolin [2020], we first bound the oscillations of the operators.

Lemma 3.4. *For all pair of functions (u, v) in $\mathrm{L}^\infty(\mathcal{X}) \times \mathrm{L}^\infty(\mathcal{Y})$, the operators Φ in* (16) *and Ψ in* (18) *satisfy the following inequalities*

$$\mathrm{osc}(\Phi(v)) \leq 2\|c\|_\infty; \tag{19}$$

$$\mathrm{osc}(\Psi(u)) \leq \frac{2\beta}{\alpha+\beta}\|c\|_\infty. \tag{20}$$

The proof is detailed in Appendix A.

With the oscillation controlled, we can establish a uniform bound.

Lemma 3.5. *For all pair of functions* (u, v) *in* $\mathrm{L}^\infty(\mathcal{X}) \times \mathrm{L}^\infty(\mathcal{Y})$*, the operators* Φ *in* (16) *and* Ψ *in* (18) *satisfy the following inequalities*

$$\|\Phi(v)\|_\infty \leq \|v\|_\infty + \|c\|_\infty; \tag{21}$$

$$\|\Psi(u)\|_\infty \leq \frac{2\beta}{\alpha+\beta}\|c\|_\infty. \tag{22}$$

The proof is detailed in Appendix A.

This uniform bound enables the application of the Banach-Alaoglu theorem, making the proof of existence possible.

Corollary 3.2. *Let* (u, v) *be a pair of functions in* $\mathrm{L}^\infty(\mathcal{X}) \times \mathrm{L}^\infty(\mathcal{Y})$*. Then, there exists another pair* (f, g) *in* $\mathrm{L}^\infty(\mathcal{X}) \times \mathrm{L}^\infty(\mathcal{Y})$ *such that*

$$\mathsf{D}(u, v) \leq \mathsf{D}(f, g), \text{ and } \|f\|_\infty, \|g\|_\infty \leq M\|c\|_\infty, \tag{23}$$

where D *is in* (10) *and* $M = \max(2, 3\frac{\beta+1}{\alpha+\beta})$.

The proof is detailed in Appendix A.

Theorem 3.1. *The supremum in problem* (10) *is attained.*

The proof is detailed in Appendix A.

3.3 Primal Attainment

Among all possible solutions to the dual problem, we deliberately defined the operators Φ and Ψ to inherently satisfy normalization conditions. This specific choice allows for a direct and simplified derivation of the primal problem's solution without additional normalization steps.

Theorem 3.2. *The infimum in problem* (5) *is attained at a unique measure* $\hat{\pi} \in \Pi(P, \star)$*. Moreover,* $\hat{\pi}$ *is absolutely continuous with respect to the product measure* PQ*, and its density can be written as*

$$\frac{\mathrm{d}\hat{\pi}}{\mathrm{d}PQ}(x, y) = e^{\frac{\hat{u}(x)}{\alpha}} e^{-\frac{\mathsf{c}(x,y)}{\alpha}} e^{\frac{\hat{v}(y)}{\alpha}}, \tag{24}$$

where $(\hat{u}, \hat{v})$ *is any pair of maximizing potentials for problem* (10) *satisfying* (15) *and* (17).

The proof is detailed in Appendix A.

The uniqueness of the primal minimizer naturally implies the uniqueness of the dual maximizer pair that simultaneously satisfies (15) and (17).

Corollary 3.3. *There exists a unique pair of functions* $(u, v) \in \mathrm{L}^\infty(\mathcal{X}) \times \mathrm{L}^\infty(\mathcal{Y})$ *satisfying* (15) *and* (17) *simultaneously.*

The proof is detailed in Appendix A.

3.4 Approximation Error Bounds

In this section, we analyze the behavior of the regularized solution when varying the parameters α and β. A crucial aspect of this regularized formulation is the implicit constraint imposed by the reference measure Q. Because of the presence of the Kullback-Leibler divergence with respect to Q, the second marginal of the solution to the regularized problem is constrained to share the support of Q. Consequently, convergence to a minimizer of the unregularized semi-relaxed problem is only possible if there exists such a minimizer whose second marginal is absolutely continuous with respect to Q. This highlights the importance of carefully selecting the reference measure Q in practice Di Marino and Gerolin [2020].

3.5 Iterative Algorithms and Convergence

3.5.1 Presentation of the Algorithms

Similar to the classical Sinkhorn algorithm, our approach relies on alternating maximization. The specific choice of the updates via Φ and Ψ is motivated by their construction: these operators automatically yield potentials that satisfy the normalization constraint. This guarantees that the associated

Algorithm 1: Unscaled iterative procedure

Let $v^{(0)}$ be in $\mathrm{L}^\infty(\mathcal{Y})$.

$$u^{(l+1)} = \Phi(v^{(l)}); \tag{25a}$$

$$v^{(l+1)} = \Psi(u^{(l+1)}), \tag{25b}$$

where Φ is in (16) and Ψ is in (18).

measure is directly a probability distribution, avoiding the need for an integral normalization factor at each iteration.

Compared to the procedure proposed by Chizat et al. [2018] in Equation 3.6, our operator Ψ includes an additional normalization constant (the first term in (18)). While this term requires computation, it provides significantly better theoretical stability and convergence guarantees, which also translates to improved robustness in numerical implementations.

Algorithm 2: Scaled iterative procedure

Let $b^{(0)} \in \mathrm{L}^\infty(\mathcal{Y})$ be equal to 1, and for all $l \in \mathbb{N}$, let the following terms of the sequence be defined as

$$a^{(l+1)} = \Xi\left(b^{(l)}\right); \tag{26a}$$

$$v^{(l+1)} = \Gamma\left(a^{(l+1)}\right), \tag{26b}$$

where the operators $\Xi : \mathrm{L}^\infty(\mathcal{Y}) \to \mathrm{L}^\infty(\mathcal{X})$ and $\Gamma : \mathrm{L}^\infty(\mathcal{X}) \to \mathrm{L}^\infty(\mathcal{Y})$ are defined by

$$\Xi(b)(x) \triangleq \frac{1}{\int_{\mathcal{Y}} b(y) e^{-\frac{\mathsf{c}(x,y)}{\alpha}} \mathrm{d}Q(y)}; \tag{27}$$

$$\Gamma(a)(y) \triangleq \left(\frac{1}{\int_{\mathcal{X}} a(x) e^{-\frac{\mathsf{c}(x,y)}{\alpha}} \mathrm{d}P(x)}\right)^{\frac{\beta}{\alpha+\beta}} \times \left(\int_{\mathcal{Y}} \left(\int_{\mathcal{X}} a(x) e^{\frac{-\mathsf{c}(x,y')}{\alpha}} \mathrm{d}P(x)\right)^{\frac{\alpha}{\alpha+\beta}} \mathrm{d}Q(y')\right)^{\frac{\beta}{\alpha}}. \tag{28}$$

3.5.2 Theoretical Guarantees

In the more classical unbalanced Sinkhorn setting with two relaxed marginals [Chizat et al., 2018, Theorem 3.11], a linear rate of convergence can be obtained by viewing each scaling update as an order-preserving map that is z-homogeneous (with $0 < z < 1$) and therefore strictly contractive in the Thompson metric [Chizat et al., 2018, Definition 3.10]. In this semi-relaxed setting, however, the scaled updates exhibit a global renormalization in (18) that destroys the required homogeneity structure in Thompson's geometry. More precisely, with the Gibbs kernel $e^{-c(x,y)/\alpha}$, the first update in (26a) is the analogue of the hard marginal projection: it is order-reversing and (-1)-homogeneous, hence non-expansive (in fact 1-Lipschitz) in projective metrics. The second update in (26b) involves a normalization factor that depends on the current iterate but is constant with respect to the spatial variable. This global rescaling is precisely what prevents the update from being z-homogeneous with $z \in (0, 1)$ in the sense required by the Thompson contraction argument. By contrast, the Hilbert projective metric is invariant under multiplication by positive scalars, so such global renormalizations are invisible at the level of distances [Lemmens and Nussbaum, 2013, Lemma 2.1(iv)]. This makes the Hilbert geometry the natural framework for the scaled scheme.

Definition 3.1 (Hilbert metric)**.** *Let* $\mathrm{L}^\infty_{++}(\mathcal{X})$ *be the subset of* $\mathrm{L}^\infty(\mathcal{X})$ *defined by*

$$\mathrm{L}^\infty_{++}(\mathcal{X}) \triangleq \{a \in \mathrm{L}^\infty(\mathcal{X}) : \inf_{x\in\mathcal{X}} a(x) > 0\} \tag{29}$$

and similarly for $\mathrm{L}^\infty_{++}(\mathcal{Y})$*. Given two functions* $f, g \in \mathrm{L}^\infty_{++}(\mathcal{Y})$*, Hilbert's projective metric is given by*

$$d_{\mathrm{H}}(f,g) \triangleq \log\left(\frac{M(f/g)}{m(f/g)}\right), \tag{30}$$

$$\text{where } M(f/g) \triangleq \sup_{y\in\mathcal{Y}} \frac{f(y)}{g(y)}, \text{ and } m(f/g) \triangleq \inf_{y\in\mathcal{Y}} \frac{f(y)}{g(y)}, \tag{31}$$

which is exactly Birkhoff's version of Hilbert's metric on cones [Lemmens and Nussbaum, 2013, Equation (2.2)].

Theorem 3.3. *Let* $\hat{u}$ *and* $\hat{v}$ *be the unique maximizers for problem* (10) *that satisfy* (15) *and* (17) *simultaneously, and define the associated scaled potentials*

$$\hat{a} = \exp\left(\frac{\hat{u}}{\alpha}\right), \qquad \hat{b} = \exp\left(\frac{\hat{v}}{\alpha}\right). \tag{32}$$

Then, the scaled iterates defined in Algorithm 2 converge to these scaled potentials at a linear rate: for all $l \geq 0$*,*

$$d_{\mathrm{H}}\left(b^{(l)}, \hat{b}\right) \leq \rho^l_{(\mathsf{c},\alpha,\beta)}\, d_{\mathrm{H}}\left(b^{(0)}, \hat{b}\right), \tag{33}$$

$$d_{\mathrm{H}}\left(a^{(l+1)}, \hat{a}\right) \leq \tanh\left(\frac{\|c\|_\infty}{\alpha}\right) \rho^l_{(\mathsf{c},\alpha,\beta)}\, d_{\mathrm{H}}\left(b^{(0)}, \hat{b}\right), \tag{34}$$

where d_{H} *is the Hilbert metric in Definition 3.1 and*

$$\rho_{(\mathsf{c},\alpha,\beta)} \triangleq \frac{\beta}{\alpha+\beta} \tanh\left(\frac{\|c\|_\infty}{\alpha}\right)^2. \tag{35}$$

The proof is detailed in Appendix A.

Having established the convergence of the scaled potentials, we can deduce the uniform convergence of the non-scaled iterates.

Corollary 3.4. *The non-scaled iterates* $(u^{(l)}, v^{(l)})$ *defined in Algorithm 1 converge uniformly to* $(\hat{u}, \hat{v})$*, the unique pair of maximizers for problem* (10) *that satisfy* (15) *and* (17) *simultaneously, at a linear rate: for all* $l \geq 0$*,*

$$\|v^{(l)} - \hat{v}\|_\infty \leq \alpha \rho^l_{(\mathsf{c},\alpha,\beta)} d_{\mathrm{H}}\left(b^{(0)}, \hat{b}\right), \tag{36}$$

$$\|u^{(l+1)} - \hat{u}\|_\infty \leq \alpha \rho^l_{(\mathsf{c},\alpha,\beta)} d_{\mathrm{H}}\left(b^{(0)}, \hat{b}\right). \tag{37}$$

The proof is detailed in Appendix A.

We now establish the linear convergence of the dual objective and the primal measures.

Corollary 3.5. *Let* $(u^{(l)}, v^{(l)})_{l\geq 0}$ *be the iterates defined by Algorithm 1. The sequence* $\left(\mathsf{D}(u^{(l)}, v^{(l)})\right)_{l\geq 0}$ *is non-decreasing and converges to* $\mathsf{P}(\hat{\pi})$ *at a linear speed:*

$$0 \;\leq\; \mathsf{P}(\hat{\pi}) - \mathsf{D}\left(u^{(l)}, v^{(l)}\right) \;\leq\; 4\alpha\, \rho^l_{(\mathsf{c},\alpha,\beta)}\, d_{\mathrm{H}}\left(b^{(0)}, \hat{b}\right), \tag{38}$$

where $\rho_{(\mathsf{c},\alpha,\beta)}$ *is defined in* (35) *and* $\hat{b} \triangleq e^{\hat{v}/\alpha}$*.*

The proof is detailed in Appendix A.

Corollary 3.6. *Let* $(u^{(l)}, v^{(l)})_{l\geq 0}$ *be the iterates defined by Algorithm 1, and let* $M \triangleq \max\left(2,\, 3\frac{\beta+1}{\alpha+\beta}\right)$*. For each* $l \geq 0$*, define the measure* $\pi^{(l)}$ *on* $\mathcal{X}\times\mathcal{Y}$ *through*

$$\frac{\mathrm{d}\pi^{(l)}}{\mathrm{d}PQ}(x,y) \;\triangleq\; \exp\left(\frac{u^{(l+1)}(x) + v^{(l)}(y) - \mathsf{c}(x,y)}{\alpha}\right). \tag{39}$$

Then $\pi^{(l)} \in \Pi(P,\star)$ *for all* $l \geq 0$*, and the sequence* $\left(\mathrm{d}\pi^{(l)}/\mathrm{d}PQ\right)$ *converges to* $\mathrm{d}\hat{\pi}/\mathrm{d}PQ$ *in* $\mathrm{L}^\infty(\mathcal{X}\times\mathcal{Y})$ *at a linear speed, that is*

$$\left\|\frac{\mathrm{d}\pi^{(l)}}{\mathrm{d}PQ} - \frac{\mathrm{d}\hat{\pi}}{\mathrm{d}PQ}\right\|_\infty \;\leq\; 2\, e^{(2M+2)\|\mathsf{c}\|_\infty/\alpha}\, \rho^l_{(\mathsf{c},\alpha,\beta)}\, d_{\mathrm{H}}\left(b^{(0)}, \hat{b}\right), \tag{40}$$

where $\rho_{(\mathsf{c},\alpha,\beta)}$ *is defined in* (35)*. In particular,* $\pi^{(l)}$ *converges to* $\hat{\pi}$ *in total variation at the same linear rate.*

The proof is detailed in Appendix A.

4 Experiments

Experimental evaluation is reserved for future extensions.

5 Conclusion and Broader Impact

We proposed a robust dual formulation and associated Sinkhorn-like algorithm for the semi-relaxed optimal transport problem. The proposed method avoids complex integral normalizations and exhibits linear convergence in both the dual potentials and the primal objective. Our analysis demonstrates that the structural characteristics of the semi-relaxed formulation permit strong theoretical guarantees without the restrictive finite space assumptions often employed. Future work will investigate the precise control of the approximation error with respect to the unregularized problem.

References

Haïm Brézis. *Functional Analysis, Sobolev Spaces and Partial Differential Equations*. Springer, New York, NY, first edition, 2011.

Lénaïc Chizat, Gabriel Peyré, Bernhard Schmitzer, and François-Xavier Vialard. Scaling algorithms for unbalanced transport problems. *Mathematics of Computation*, 87(314):2563–2609, Feb. 2018. doi: 10.1090/mcom/3303.

Marco Cuturi. Sinkhorn distances: Lightspeed computation of optimal transportation distances. In *Proceedings of the International Conference on Neural Information Processing Systems (NeurIPS)*, volume 2, pages 2292–2300, Dec. 2013.

Simone Di Marino and Augusto Gerolin. Optimal transport losses and sinkhorn algorithm with general convex regularization. *arXiv preprint arXiv:2007.00976*, Jul. 2020.

Bas Lemmens and Roger D. Nussbaum. Birkhoff's version of hilbert's metric and its applications in analysis. *arXiv: Metric Geometry*, 2013. URL `https://api.semanticscholar.org/CorpusID:119321129`.

Roger D. Nussbaum. Fixed point theorems and denjoy–wolff theorems for hilbert's projective metric in infinite dimensions. *Topological Methods in Nonlinear Analysis*, 29(2):199–249, 2007.

Walter Rudin. *Real and Complex Analysis*. McGraw-Hill, New York, NY, USA, third edition, 1987.

Constantin Zalinescu. *Convex analysis in general vector spaces*. World Scientific, Singapore, first edition, 2002.

A Technical appendices and supplementary material

This appendix contains the proofs of the theoretical results presented in the main text.

A.1 Proof of Lemma 3.1

Proof: Let $\pi \in \Pi(P, \star)$ be any feasible measure for problem (5); in particular π is absolutely continuous with respect to the product measure PQ and $\pi^{\mathcal{X}}_{\#} = P$. Let $(u, v) \in \mathrm{L}^\infty(\mathcal{X}) \times \mathrm{L}^\infty(\mathcal{Y})$ be arbitrary potentials. The Fenchel-Young inequality for the function $a \mapsto \alpha(a\log(a) - a + 1)$ states

$$\alpha(a\log(a) - a + 1) \geq at - \alpha(e^{t/\alpha} - 1) \geq at - e^{t/\alpha}, \tag{41}$$

for all $t \in \mathbb{R}$ and $a > 0$. Let (x, y) be in $\mathcal{X} \times \mathcal{Y}$; applying the previous inequality with $a = \frac{\mathrm{d}\pi}{\mathrm{d}PQ}(x, y)$ and $t = u(x) + v(y) - \mathsf{c}(x, y)$ and integrating with respect to PQ yields

$$\begin{aligned}\alpha D_{\mathrm{KL}}(\pi||PQ) \geq &\int_{\mathcal{X}\times\mathcal{Y}} (u(x) + v(y) - \mathsf{c}(x, y))\mathrm{d}\pi(x, y) \\ &-\alpha \log\left(\int_{\mathcal{X}\times\mathcal{Y}} e^{\frac{u(x)+v(y)-\mathsf{c}(x,y)}{\alpha}} \mathrm{d}PQ(x, y)\right).\end{aligned} \tag{42}$$

Similarly, the following inequality can be deduced from (41)

$$\beta D_{\mathrm{KL}}(\pi^{\mathcal{Y}}_{\#}||Q) \geq \int_{\mathcal{Y}} -v(y)\mathrm{d}\pi^{\mathcal{Y}}_{\#}(y) - \beta \log\left(\int_{\mathcal{Y}} e^{-\frac{v(y)}{\beta}}\mathrm{d}Q(y)\right). \tag{43}$$

Addition of the primal cost term $\int \mathsf{c}(x,y)\mathrm{d}\pi(x,y)$ to the sum of these inequalities leads to

$$\begin{aligned}\mathsf{P}(\pi) \geq \int_{\mathcal{X}\times\mathcal{Y}} (u(x)+v(y)-\mathsf{c}(x,y))\mathrm{d}\pi(x,y) + \int_{\mathcal{X}\times\mathcal{Y}} \mathsf{c}(x,y)\mathrm{d}\pi(x,y) - \int_{\mathcal{Y}} v(y)\mathrm{d}\pi^{\mathcal{Y}}_{\#}(y)\\ -\alpha\log\left(\int_{\mathcal{X}\times\mathcal{Y}} e^{\frac{u(x)+v(y)-\mathsf{c}(x,y)}{\alpha}}\mathrm{d}PQ(x,y)\right) - \beta\log\left(\int_{\mathcal{Y}} e^{-\frac{v(y)}{\beta}}\mathrm{d}Q(y)\right).\end{aligned} \tag{44}$$

The definition of the marginals and the constraint $\pi \in \Pi(P,\star)$ imply the identity

$$\int_{\mathcal{X}\times\mathcal{Y}} (u(x)+v(y))\mathrm{d}\pi(x,y) = \int_{\mathcal{X}} u(x)\mathrm{d}P(x) + \int_{\mathcal{Y}} v(y)\mathrm{d}\pi^{\mathcal{Y}}_{\#}(y). \tag{45}$$

Substitution of this identity into the previous expression results in the cancellation of terms involving c and v. The remaining terms satisfy

$$\begin{aligned}\mathsf{P}(\pi) \geq \int_{\mathcal{X}} u(x)\mathrm{d}P(x) - \alpha\log\left(\int_{\mathcal{X}\times\mathcal{Y}} e^{\frac{u(x)+v(y)-\mathsf{c}(x,y)}{\alpha}}\mathrm{d}PQ(x,y)\right)\\ -\beta\log\left(\int_{\mathcal{Y}} e^{-\frac{v(y)}{\beta}}\mathrm{d}Q(y)\right),\end{aligned} \tag{46}$$

where the right-hand term is exactly $\mathsf{D}(u,v)$. This inequality holds for all u, v and all feasible π, thus confirming $\sup \mathsf{D}(u,v) \leq \inf \mathsf{P}(\pi)$. Regarding the finiteness of the two values, note that from the fact that c is bounded and PQ is in $\Pi(P,Q)$ and satisfies $D_{\mathrm{KL}}(PQ||PQ) = 0$, it holds that $\mathsf{P}(PQ) < +\infty$. ■

A.2 Proof of Lemma 3.2

Proof: Let u and v be two functions in $\mathrm{L}^\infty(\mathcal{X})$ and $\mathrm{L}^\infty(\mathcal{Y})$ respectively, and M, N be two real numbers. From (10), it follows that

$$\begin{aligned}\mathsf{D}(u+M,v) &= \int_{\mathcal{X}} u(x)\mathrm{d}P(x) + M - \beta\log\left(\int_{\mathcal{Y}} e^{-\frac{v(y)}{\beta}}\mathrm{d}Q(y)\right)\\ &\quad -\alpha\log\left(e^{\frac{M}{\alpha}}\int_{\mathcal{X}\times\mathcal{Y}} e^{\frac{u(x)+v(y)-\mathsf{c}(x,y)}{\alpha}}\mathrm{d}PQ(x,y)\right) &(47)\\ &= \int_{\mathcal{X}} u(x)\mathrm{d}P(x) - \beta\log\left(\int_{\mathcal{Y}} e^{-\frac{v(y)}{\beta}}\mathrm{d}Q(y)\right)\\ &\quad -\alpha\log\left(\int_{\mathcal{X}\times\mathcal{Y}} e^{\frac{u(x)+v(y)-\mathsf{c}(x,y)}{\alpha}}\mathrm{d}PQ(x,y)\right). &(48)\\ &= \mathsf{D}(u,v), &(49)\end{aligned}$$

which proves the first equality in (12). The same computation can be done to prove the second one, as follows

$$\begin{aligned}\mathsf{D}(u,v+N) &= \int_{\mathcal{X}} u(x)\mathrm{d}P(x) - \beta\log\left(e^{-\frac{N}{\beta}}\int_{\mathcal{Y}} e^{-\frac{v(y)}{\beta}}\mathrm{d}Q(y)\right)\\ &\quad -\alpha\log\left(e^{\frac{N}{\alpha}}\int_{\mathcal{X}\times\mathcal{Y}} e^{\frac{u(x)+v(y)-\mathsf{c}(x,y)}{\alpha}}\mathrm{d}PQ(x,y)\right) &(50)\\ &= \int_{\mathcal{X}} u(x)\mathrm{d}P(x) - \beta\log\left(\int_{\mathcal{Y}} e^{-\frac{v(y)}{\beta}}\mathrm{d}Q(y)\right) + N\\ &\quad -\alpha\log\left(\int_{\mathcal{X}\times\mathcal{Y}} e^{\frac{u(x)+v(y)-\mathsf{c}(x,y)}{\alpha}}\mathrm{d}PQ(x,y)\right) - N. &(51)\\ &= \mathsf{D}(u,v), &(52)\end{aligned}$$

which concludes the proof of (12). ■

A.3 Proof of Lemma 3.3

Proof: The proof is split into two parts, establishing (13) and (14) respectively. Let (u, v) be in $L^\infty(\mathcal{X}) \times L^\infty(\mathcal{Y})$, let a be in $L^\infty(\mathcal{X})$, and let b be in $L^\infty(\mathcal{Y})$. Since the functions u, v, a, b are bounded, and $\exp$ is continuous, all subsequent differentiations under the integral sign are justified by the dominated convergence theorem [Rudin, 1987, Theorem 1.34]. Consider first the derivative with respect to u. The first term of the functional D in (10) is linear with respect to u and conributes

$$\frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} \int_{\mathcal{X}} (u(x) + ta(x))\,\mathrm{d}P(x) = \int_{\mathcal{X}} a(x)\,\mathrm{d}P(x), \tag{53}$$

while the term $-\beta \log\big(\int_{\mathcal{Y}} e^{-v/\beta}\,\mathrm{d}Q\big)$ does not depend on u and therefore yields zero when differentiated with respect to u. For the coupling term, differentiation gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} Z(u{+}ta{,}v) = \frac{1}{\alpha}\int_{\mathcal{X}\times\mathcal{Y}} a(x)\,\exp\!\left(\frac{u(x)+v(y)-\mathsf{D}(x,y)}{\alpha}\right)\,\mathrm{d}PQ(x,y). \tag{54}$$

Applying the chain rule for the logarithm yields

$$\begin{aligned}&\frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} \big(-\alpha\log Z(u+ta,v)\big)\\ &= -\frac{1}{Z(u,v)}\int_{\mathcal{X}\times\mathcal{Y}} a(x)\,\exp\!\left(\frac{u(x)+v(y)-\mathsf{D}(x,y)}{\alpha}\right)\,\mathrm{d}PQ(x,y).\end{aligned} \tag{55}$$

By Fubini's theorem [Rudin, 1987, Theorem 8.8], this can be rewritten as

$$\begin{aligned}&\frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} \big(-\alpha\log Z(u+ta,v)\big)\\ &= -\int_{\mathcal{X}} a(x)\left(\frac{1}{Z(u,v)}\int_{\mathcal{Y}} \exp\!\left(\frac{u(x)+v(y)-\mathsf{D}(x,y)}{\alpha}\right)\,\mathrm{d}Q(y)\right)\mathrm{d}P(x).\end{aligned} \tag{56}$$

Combining (53) and (56) concludes the proof of (13). The differentiation with respect to v is handled analogously. The term $\int_{\mathcal{X}} u\,\mathrm{d}P$ in (10) does not depend on v. For the logarithmic term, differentiation yields

$$\frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0}\left(-\beta\log\int_{\mathcal{Y}} e^{-(v(y)+tb(y))/\beta}\,\mathrm{d}Q(y)\right) = \frac{1}{\int_{\mathcal{Y}} e^{-v/\beta}\mathrm{d}Q}\int_{\mathcal{Y}} b(y)e^{-v(y)/\beta}\mathrm{d}Q(y). \tag{57}$$

The coupling term is handled analogously, by symmetry with respect to the variables u and v, and the result is as follows

$$\begin{aligned}&\frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} \big(-\alpha\log Z(u,v+tb)\big)\\ &= -\int_{\mathcal{Y}} b(y)\left(\frac{1}{Z(u,v)}\int_{\mathcal{X}} \exp\!\left(\frac{u(x)+v(y)-\mathsf{D}(x,y)}{\alpha}\right)\,\mathrm{d}P(x)\right)\mathrm{d}Q(y).\end{aligned} \tag{58}$$

Summing (57) and (58) concludes the proof of (14). ■

A.4 Proof of Corollary 3.1

Proof: The functional D is differentiable and concave on $L^\infty(\mathcal{X}) \times L^\infty(\mathcal{Y})$, and in particular concave with respect to each variable when the other one is fixed. As a consequence, a sufficient condition for optimality is the vanishing of the first-order derivative [Zalinescu, 2002, Chapter 2.4].

Assume that $\hat{v} \in L^\infty(\mathcal{Y})$ is fixed and that $\hat{u}$ satisfies (15). By Lemma 3.3, the first-order derivative of the map $u \mapsto \mathsf{D}(u, \hat{v})$ is given by (13). Injecting (15) into this expression shows that the derivative vanishes identically. By concavity, $\hat{u}$ therefore maximizes $u \mapsto \mathsf{D}(u, \hat{v})$.

Similarly, assume that $\hat{u} \in L^\infty(\mathcal{X})$ is fixed and that $\hat{v}$ satisfies (17). Using again Lemma 3.3, the derivative of $v \mapsto \mathsf{D}(\hat{u}, v)$ is given by (14). Injecting (17) into this expression yields the cancellation of the two terms and thus the vanishing of the derivative. By concavity, $\hat{v}$ maximizes $v \mapsto \mathsf{D}(\hat{u}, v)$.

Finally, if (15) and (17) are satisfied simultaneously, then both partial derivatives vanish. Since $(u, v) \mapsto \mathsf{D}(u, v)$ is jointly concave, $(\hat{u}, \hat{v})$ is a global maximizer of D. ■

A.5 Proof of Lemma 3.4

Proof: The proof is divided in two parts, establishing (19) and (20) respectively. Let v be a function in $\mathrm{L}^\infty(\mathcal{Y})$ and (x, y) be in $\mathcal{X} \times \mathcal{Y}$; the following inequalities are satisfied

$$-\|c\|_\infty \leq \quad -\mathsf{c}(x,y) \quad \leq \|c\|_\infty \tag{59}$$

$$e^{\frac{-\|c\|_\infty}{\alpha}} e^{\frac{v(y)}{\alpha}} \leq \quad e^{\frac{v(y)-\mathsf{c}(x,y)}{\alpha}} \quad \leq e^{\frac{\|c\|_\infty}{\alpha}} e^{\frac{v(y)}{\alpha}} \tag{60}$$

$$e^{\frac{-\|c\|_\infty}{\alpha}} \int_{\mathcal{Y}} e^{\frac{v(y)}{\alpha}} \mathrm{d}Q(y) \leq \int_{\mathcal{Y}} e^{\frac{v(y)-\mathsf{c}(x,y)}{\alpha}} \mathrm{d}Q(y) \leq e^{\frac{\|c\|_\infty}{\alpha}} \int_{\mathcal{Y}} e^{\frac{v(y)}{\alpha}} \mathrm{d}Q(y). \tag{61}$$

Taking the logarithm of the previous inequality and multiplying by $-\alpha$ yields

$$-\alpha \log\left(\int_{\mathcal{Y}} e^{\frac{v(y)}{\alpha}} \mathrm{d}Q(y)\right) - \|c\|_\infty \leq \Phi(v)(x) \leq -\alpha \log\left(\int_{\mathcal{Y}} e^{\frac{v(y)}{\alpha}} \mathrm{d}Q(y)\right) + \|c\|_\infty, \tag{62}$$

from which it can be deduced that $\mathrm{osc}(\Phi(v)) \leq 2\|c\|_\infty$, hence concluding the proof of (19).

To establish (20), let Ψ_1 and Ψ_2 denote the first and second terms of Ψ in (18), respectively. Consider a function $u \in \mathrm{L}^\infty(\mathcal{X})$. The oscillation of the constant part $\Psi_1(u)$ is zero, which implies the identity $\mathrm{osc}(\Psi(u)) = \mathrm{osc}(\Psi_2(u))$. Apart from the probability measure P and the multiplicative factor $\frac{\beta}{\alpha+\beta}$, the functional form of Ψ_2 is symmetric to the definition of Φ in (16) . The proof of (19) establishes that $\mathrm{osc}(\Phi(v)) \leq 2\|c\|_\infty$. Adoption of an identical reasoning for Ψ_2 with its specific scaling leads to the following bound

$$\mathrm{osc}(\Psi(u)) \leq \frac{2\beta}{\alpha+\beta} \|c\|_\infty, \tag{63}$$

which concludes the proof of (20). ■

A.6 Proof of Lemma 3.5

Proof: The proof is divided in two parts, establishing (21) and (22) respectively. Let v be a function in $\mathrm{L}^\infty(\mathcal{Y})$ and (x, y) be in $\mathcal{X} \times \mathcal{Y}$; the following inequalities are satisfied

$$-\frac{\|v\|_\infty + \|c\|_\infty}{\alpha} \leq \frac{v(y) - \mathsf{c}(x,y)}{\alpha} \leq \frac{\|v\|_\infty + \|c\|_\infty}{\alpha} \tag{64}$$

$$e^{-\frac{\|v\|_\infty + \|c\|_\infty}{\alpha}} \leq e^{\frac{v(y)-\mathsf{c}(x,y)}{\alpha}} \leq e^{\frac{\|v\|_\infty + \|c\|_\infty}{\alpha}}. \tag{65}$$

Succesively integrating the previous inequalities with respect to Q, taking the logarithm and multiplying by $-\alpha$ concludes the proof of (21).

To establish (22), let u be in $\mathrm{L}^\infty(\mathcal{X})$ and note that the definition of Ψ in (18) implies

$$\int_{\mathcal{Y}} e^{-\Psi(u)(y)/\beta} \mathrm{d}Q(y) = 1. \tag{66}$$

As the measure Q has total mass 1, the previous equality implies

$$\inf_{y \in \mathcal{Y}} e^{-\Psi(u)(y)/\beta} \leq 1, \text{ and } \sup_{y \in \mathcal{Y}} e^{-\Psi(u)(y)/\beta} \geq 1. \tag{67}$$

Monotonicity of the logarithm function yields

$$\sup \Psi(u) \geq 0 \quad \text{and} \quad \inf \Psi(u) \leq 0. \tag{68}$$

The interval $[\inf \Psi(u), \sup \Psi(u)]$ contains the origin. The distance from the origin to any point within a closed interval is bounded through the total interval length. The following inequalities result from this inclusion

$$\sup \Psi(u) \leq \mathrm{osc}(\Psi(u)) \quad \text{and} \quad -\inf \Psi(u) \leq \mathrm{osc}(\Psi(u)). \tag{69}$$

The definition $\|\Psi(u)\|_\infty = \max(\sup \Psi(u), -\inf \Psi(u))$ establishes the bound $\|\Psi(u)\|_\infty \leq \mathrm{osc}(\Psi(u))$. Since the oscillation bound in (20) is established as $\frac{2\beta}{\alpha+\beta}\|c\|_\infty$, this concludes the proof of (22). ■

A.7 Proof of Corollary 3.2

Proof: Let (u, v) be a pair of functions in $\mathrm{L}^\infty(\mathcal{X}) \times \mathrm{L}^\infty(\mathcal{Y})$. From Lemma 3.4, the function $\Phi(u)$ satisfies $\mathrm{osc}(\Phi(u)) \leq 2\|c\|_\infty$, where Φ is in (16). Thus, there exists a constant $A \in \mathbb{R}$ such that $g \triangleq \Phi(u) + A$ satisfies

$$\|g\|_\infty \leq 2\|c\|_\infty. \tag{70}$$

From Lemma 3.2, it holds that $\mathsf{D}(u, \Phi(u)) = \mathsf{D}(u, g)$, so it follows from Corollary 3.1 that

$$\mathsf{D}(u, v) \leq \mathsf{D}(u, \phi(u)) = \mathsf{D}(u, g). \tag{71}$$

Now let $f \triangleq \Psi(g)$, where Ψ is in (18). From Corollary 3.1, the function f satisfies

$$\mathsf{D}(u, g) \leq \mathsf{D}(f, g). \tag{72}$$

Moreover, from Lemma 3.5, f also satisfies

$$\|f\|_\infty \leq \frac{\beta + 1}{\alpha + \beta}(\|g\|_\infty + \|c\|_\infty). \tag{73}$$

Hence, combining the previous inequality with (70) yields

$$\|f\|_\infty \leq 3\frac{\beta + 1}{\alpha + \beta}\|c\|_\infty, \tag{74}$$

The combination of (70) and (74) ensures that $\|f\|_\infty, \|g\|_\infty \leq M\|c\|_\infty$ with $M = \max(2, 3\frac{\beta+1}{\alpha+\beta})$. This results in a pair (f, g) satisfying $\mathsf{D}(u, v) \leq \mathsf{D}(f, g)$, from (71) and (72), which concludes the proof. ■

A.8 Proof of Theorem 3.1

Proof: Let $(u_n) \subset \mathrm{L}^\infty(\mathcal{X})$ and $(v_n) \subset \mathrm{L}^\infty(\mathcal{Y})$ be maximizing sequences. It follows from Corollary 3.2 that (u_n) and (v_n) can be chosen such that for all n, it holds that $\|u_n\|_\infty, \|v_n\|_\infty \leq M\|c\|_\infty$, with $M = \max(2, 3\frac{\beta+1}{\alpha+\beta})$. Application of the Banach-Alaoglu theorem [Brézis, 2011, Theorem 3.16] yields existence of subsequences $(u_{n_k})_k$ and $(v_{n_k})_k$ such that

$$u_n \rightharpoonup^* \overline{u} \text{ in } \mathrm{L}^\infty(\mathcal{X}), \text{ and } v_n \rightharpoonup^* \overline{v} \text{ in } \mathrm{L}^\infty(\mathcal{Y}). \tag{75}$$

In particular, the following weak$*$ convergence also holds in $\mathrm{L}^\infty(\mathcal{X} \times \mathcal{Y})$

$$u_n + v_n - \mathsf{c} \rightharpoonup^* \overline{u} + \overline{v} - \mathsf{c}. \tag{76}$$

Thus, it follows from the convexity of $\exp$ and the weak$*$ lower semi-continuity of the integral functionals in the dual objective in (10) ([Brézis, 2011, Theorem 3.7]) that

$$\liminf_{k\to\infty} \int_{\mathcal{Y}} e^{-\frac{v_{n_k}(y)}{\beta}} \mathrm{d}Q(y) \geq \int_{\mathcal{Y}} e^{-\frac{\overline{v}(y)}{\beta}} \mathrm{d}Q(y); \text{ and} \tag{77}$$

$$\liminf_{k\to\infty} \int_{\mathcal{X}\times\mathcal{Y}} e^{\frac{u_{n_k}(x)+v_{n_k}(y)-\mathsf{c}(x,y)}{\alpha}} \mathrm{d}PQ(x,y) \geq \int_{\mathcal{X}\times\mathcal{Y}} e^{\frac{\overline{u}(x)+\overline{v}(y)-\mathsf{c}(x,y)}{\alpha}} \mathrm{d}PQ(x,y). \tag{78}$$

From the continuity and monotonicity of $\log$ on $(0, \infty)$ and the two previous inequalities, it follows that

$$\liminf_{k\to\infty} \log\left(\int_{\mathcal{Y}} e^{-\frac{v_{n_k}(y)}{\beta}} \mathrm{d}Q(y)\right) \geq \log\left(\int_{\mathcal{Y}} e^{-\frac{\overline{v}(y)}{\beta}} \mathrm{d}Q(y)\right); \text{ and} \tag{79}$$

$$\liminf_{k\to\infty} \log\left(\int_{\mathcal{X}\times\mathcal{Y}} e^{\frac{u_{n_k}(x)+v_{n_k}(y)-\mathsf{c}(x,y)}{\alpha}} \mathrm{d}PQ(x,y)\right) \geq \log\left(\int_{\mathcal{X}\times\mathcal{Y}} e^{\frac{\overline{u}(x)+\overline{v}(y)-\mathsf{c}(x,y)}{\alpha}} \mathrm{d}PQ(x,y)\right). \tag{80}$$

From which it can be deduced that

$$\begin{aligned}
&\sup_{u\in \mathrm{L}^\infty(\mathcal{X}), v\in \mathrm{L}^\infty(\mathcal{Y})} \mathsf{D}(u,v) \\
&= \lim_{k\to\infty}\Bigg[\int_{\mathcal{X}} u_{n_k}(x)\mathrm{d}P(x) - \beta\log\Bigg(\int_{\mathcal{Y}} e^{-\frac{v_{n_k}(y)}{\beta}}\mathrm{d}Q(y)\Bigg) \\
&\quad -\alpha\log\Bigg(\int_{\mathcal{X}\times\mathcal{Y}} e^{\frac{u_{n_k}(x)+v_{n_k}(y)-\mathsf{c}(x,y)}{\alpha}}\mathrm{d}PQ(x,y)\Bigg)\Bigg] \qquad (81)\\
&\leq \lim_{k\to\infty}\left[\int_{\mathcal{X}} u_{n_k}(x)\mathrm{d}P(x)\right] - \beta\liminf_{k\to\infty}\left[\log\left(\int_{\mathcal{Y}} e^{-\frac{v_{n_k}(y)}{\beta}}\mathrm{d}Q(y)\right)\right] \\
&\quad -\alpha\liminf_{k\to\infty}\left[\log\Bigg(\int_{\mathcal{X}\times\mathcal{Y}} e^{\frac{u_{n_k}(x)+v_{n_k}(y)-\mathsf{c}(x,y)}{\alpha}}\mathrm{d}PQ(x,y)\Bigg)\right]. \qquad (82)
\end{aligned}$$

It follows from the weak$*$ convergence of (u_{n_k}) to $\overline{u}$ that

$$\lim_{k\to\infty}\left[\int_{\mathcal{X}} u_{n_k}(x)\mathrm{d}P(x)\right] = \int_{\mathcal{X}} \overline{u}(x)\mathrm{d}P(x). \qquad (83)$$

Combining (79), (80), (82) and the previous equality yields

$$\begin{aligned}
\sup_{u\in \mathrm{L}^\infty(\mathcal{X}), v\in \mathrm{L}^\infty(\mathcal{Y})} \mathsf{D}(u,v) \leq &\int_{\mathcal{X}} \overline{u}(x)\mathrm{d}P(x) - \beta\log\left(\int_{\mathcal{Y}} e^{-\frac{\overline{v}(y)}{\beta}}\mathrm{d}Q(y)\right) \\
&-\alpha\log\left(\int_{\mathcal{X}\times\mathcal{Y}} e^{\frac{\overline{u}(x)+\overline{v}(y)-\mathsf{c}(x,y)}{\alpha}}\mathrm{d}PQ(x,y)\right), \qquad (84)
\end{aligned}$$

or equivalently $\sup_{u\in \mathrm{L}^\infty(\mathcal{X}), v\in \mathrm{L}^\infty(\mathcal{Y})} \mathsf{D}(u,v) \leq \mathsf{D}(\overline{u},\overline{v})$, which concludes the proof. ■

A.9 Proof of Theorem 3.2

Proof: Let $(\hat{u},\hat{v}) \in \mathrm{L}^\infty(\mathcal{X}) \times \mathrm{L}^\infty(\mathcal{Y})$ be a pair of maximizing potentials for problem (10) satisfying (15) and (17). Define $\hat{\pi}$ as in (24). Let $\hat{\rho}$ denote the density $\frac{\mathrm{d}\hat{\pi}}{\mathrm{d}PQ}$ and $\hat{s}$ denote the density $\frac{\mathrm{d}\hat{\pi}^{\mathcal{Y}}_{\#}}{\mathrm{d}Q}$. Condition (15) ensures that for all $x \in \mathcal{X}$, it holds that

$$\int_{\mathcal{Y}} \hat{\rho}(x,y)\mathrm{d}Q(y) = 1. \qquad (85)$$

Thus, for any integrable function $f \in \mathrm{L}^1(\mathcal{X})$, Fubini's theorem [Rudin, 1987, Theorem 8.8] yields

$$\int_{\mathcal{X}\times\mathcal{Y}} f(x)\mathrm{d}\hat{\pi}(x,y) = \int_{\mathcal{X}} f(x)\left(\int_{\mathcal{Y}} \hat{\rho}(x,y)\mathrm{d}Q(y)\right)\mathrm{d}P(x) = \int_{\mathcal{X}} f(x)\mathrm{d}P(x). \qquad (86)$$

Hence, the measure $\hat{\pi}$ is a probability measure with first marginal $\hat{\pi}^{\mathcal{X}}_{\#} = P$, which implies $\hat{\pi} \in \Pi(P,\star)$. Besides, the optimality condition (17) implies

$$\int_{\mathcal{Y}} e^{-\hat{v}(y)/\beta}\mathrm{d}Q(y) = 1, \text{ and } \hat{s} = e^{-\hat{v}/\beta}. \qquad (87)$$

From (85) and (87), the dual objective $\mathsf{D}(\hat{u},\hat{v})$, with D in (10), can be computed as follows

$$\begin{aligned}
\mathsf{D}(\hat{u},\hat{v}) = &\int_{\mathcal{X}} \hat{u}(x)\mathrm{d}P(x) - \beta\log\left(\int_{\mathcal{Y}} e^{-\frac{\hat{v}(y)}{\beta}}\mathrm{d}Q(y)\right) \\
&-\alpha\log\left(\int_{\mathcal{X}\times\mathcal{Y}} e^{\frac{\hat{u}(x)+\hat{v}(y)-\mathsf{c}(x,y)}{\alpha}}\mathrm{d}PQ(x,y)\right) \qquad (88)\\
= &\int_{\mathcal{X}} \hat{u}(x)\mathrm{d}P(x) - \beta\log(1) - \alpha\log(1) = \int_{\mathcal{X}} \hat{u}(x)\mathrm{d}P(x). \qquad (89)
\end{aligned}$$

From (5), the primal objective value at $\hat{\pi}$ is defined via

$$\begin{aligned}
\mathsf{P}(\hat{\pi}) = &\int_{\mathcal{X}\times\mathcal{Y}} (\mathsf{c}(x,y)\hat{\rho}(x,y) + \alpha(\hat{\rho}(x,y)\log(\hat{\rho}(x,y)) - \hat{\rho}(x,y) + 1))\mathrm{d}PQ(x,y) \\
&+\beta\int_{\mathcal{Y}} (\hat{s}(y)\log(\hat{s}(y)) - \hat{s}(y) + 1)\mathrm{d}Q(y). \qquad (90)
\end{aligned}$$

For all points $(x, y) \in \mathcal{X} \times \mathcal{Y}$, since $\hat{\rho}(x, y) = e^{\frac{\hat{u}(x)+\hat{v}(y)-\mathsf{c}(x,y)}{\alpha}}$, the pointwise Fenchel-Young equality [Zalinescu, 2002, Theorem 2.3.1] for the function $t \mapsto \alpha(t \log(t) - t + 1)$ states

$$\alpha(\hat{\rho}(x,y)\log(\hat{\rho}(x,y))-\hat{\rho}(x,y)+1)=\hat{\rho}(x,y)(\hat{u}(x)+\hat{v}(y)-\mathsf{c}(x,y))-\alpha(\hat{\rho}(x,y)-1). \tag{91}$$

Substitution into (90) leads to

$$\begin{aligned}\mathsf{P}(\hat{\pi}) = &\int_{\mathcal{X}\times\mathcal{Y}} \hat{\rho}(x, y)(\hat{u}(x) + \hat{v}(y))\mathrm{d}PQ(x, y) - \int_{\mathcal{X}\times\mathcal{Y}} \alpha(\hat{\rho}(x, y) - 1)\mathrm{d}PQ(x, y) \\ &+\beta \int_{\mathcal{Y}} (\hat{s}(y) \log(\hat{s}(y)) - \hat{s}(y) + 1)\mathrm{d}Q(y).\end{aligned} \tag{92}$$

As $\hat{\rho}$ is a probability density, the intergral term $\int_{\mathcal{X}\times\mathcal{Y}}(\hat{\rho}(x, y) - 1)\mathrm{d}PQ(x, y)$ vanishes. Equation (87) provides the relation $\log(\hat{s}) = -\hat{v}/\beta$. Pointwise Fenchel-Young equality [Zalinescu, 2002, Theorem 2.3.1] for the second entropic term in (90) implies

$$\beta(\hat{s}(y) \log(\hat{s}(y)) - \hat{s}(y) + 1) = -\hat{s}(y)\hat{v}(y) - \beta(\hat{s}(y) - 1). \tag{93}$$

Linearity of the integral and Fubini's theorem [Rudin, 1987, Theorem 8.8] yield

$$\begin{aligned}\mathsf{P}(\hat{\pi}) = &\int_{\mathcal{X}} \hat{u}(x) \left(\int_{\mathcal{Y}} \hat{\rho}(x, y)\mathrm{d}Q(y)\right) \mathrm{d}P(x) + \int_{\mathcal{Y}} \hat{v}(y) \left(\int_{\mathcal{X}} \hat{\rho}(x, y)\mathrm{d}P(x)\right) \mathrm{d}Q(y) \\ &- \int_{\mathcal{Y}} \hat{v}(y)\hat{s}(y)\mathrm{d}Q(y) - \beta \int_{\mathcal{Y}} (\hat{s}(y) - 1)\mathrm{d}Q(y).\end{aligned} \tag{94}$$

Definitions of the marginals $\hat{\pi}^{\mathcal{X}}_{\#}$ and $\hat{\pi}^{\mathcal{Y}}_{\#}$ coupled with identities (85) and (87) lead to

$$\mathsf{P}(\hat{\pi}) = \int_{\mathcal{X}} \hat{u}(x)\mathrm{d}P(x)+\int_{\mathcal{Y}} \hat{v}(y)\hat{s}(y)\mathrm{d}Q(y)-\int_{\mathcal{Y}} \hat{v}(y)\hat{s}(y)\mathrm{d}Q(y)-\beta(1-1) \tag{95}$$

$$= \int_{\mathcal{X}} \hat{u}(x)\mathrm{d}P(x) = D(\hat{u}, \hat{v}), \tag{96}$$

where the last equality follows from (89). Therefore, from Lemma 3.1, strong duality $\mathsf{P}(\hat{\pi}) = \mathsf{D}(\hat{u}, \hat{v})$ holds and $\hat{\pi}$ is a minimizer for P. Uniqueness of the minimizer $\hat{\pi}$ follows from the strict convexity of the Kullback-Leibler divergence on its support. ■

A.10 Proof of Corollary 3.3

Proof: Let (u_1, v_1) and (u_2, v_2) satisfy (15) and (17) simultaneously, that is

$$u_1 = \Phi(v_1), \qquad v_1 = \Psi(u_1), \tag{97}$$

$$u_2 = \Phi(v_2), \qquad v_2 = \Psi(u_2), \tag{98}$$

Associate to each pair the coupling defined as follows

$$\pi_1(\mathrm{d}x, \mathrm{d}y) \triangleq \exp\left(\frac{u_1(x) + v_1(y) - \mathsf{c}(x, y)}{\alpha}\right) P(\mathrm{d}x)\, Q(\mathrm{d}y), \tag{99}$$

$$\pi_2(\mathrm{d}x, \mathrm{d}y) \triangleq \exp\left(\frac{u_2(x) + v_2(y) - \mathsf{c}(x, y)}{\alpha}\right) P(\mathrm{d}x)\, Q(\mathrm{d}y). \tag{100}$$

From Theorem 3.2, (97) and (98), π_1 and π_2 are primal minimizers. Uniqueness of the primal minimizer (Theorem 3.2) yields

$$\pi_1 = \pi_2, \tag{101}$$

which, together with (99) and (100), implies

$$\exp\left(\frac{u_1(x) + v_1(y) - \mathsf{c}(x, y)}{\alpha}\right) \overset{a.s.}{\underset{PQ}{=}} \exp\left(\frac{u_2(x) + v_2(y) - \mathsf{c}(x, y)}{\alpha}\right). \tag{102}$$

Taking logarithms in the previous equality gives

$$u_1(x) + v_1(y) \overset{a.s.}{\underset{PQ}{=}} u_2(x) + v_2(y). \tag{103}$$

Fix y_0 outside a Q-negligible set such that (103) holds for P-a.e. x. Then, for P-a.e. x,

$$u_1(x) - u_2(x) = v_2(y_0) - v_1(y_0) \overset{a.s.}{\underset{P}{=}} K, \tag{104}$$

for the constant $K \triangleq v_2(y_0) - v_1(y_0)$. Insert (104) into (103) to obtain, for Q-a.e. y,

$$v_2(y) - v_1(y) = K. \tag{105}$$

Finally, combining (105) with the normalization (66), which follows from (97) and (98), yields

$$1 = \int_{\mathcal{Y}} \exp\left(-\frac{v_2(y)}{\beta}\right) \mathrm{d}Q(y) \tag{106}$$

$$= \int_{\mathcal{Y}} \exp\left(-\frac{v_1(y) + K}{\beta}\right) \mathrm{d}Q(y) \tag{107}$$

$$= e^{-K/\beta} \int_{\mathcal{Y}} \exp\left(-\frac{v_1(y)}{\beta}\right) \mathrm{d}Q(y) = e^{-K/\beta}, \tag{108}$$

which implies $K = 0$, hence (104) and (105) yield

$$u_1 \overset{a.s.}{\underset{P}{=}} u_2, \qquad v_1 \overset{a.s.}{\underset{Q}{=}} v_2. \tag{109}$$

This proves uniqueness of the pair satisfying (15) and (17). ■

A.11 Proof of Theorem 3.3

Proof: The proof is organized as follows. First, two positive linear operators are introduced and their projective diameters shown to be are bounded. Theorem 2.9 in Lemmens and Nussbaum [2013] then yields strict contractions in Hilbert's metric. Next, the operators Ξ and Γ are shown to be Lipschitz, and the one step map $\Gamma \circ \Xi$ is shown to be a strict contraction. Finally, the normalization built into Γ identifies a complete slice of the positive cone, and the Banach fixed point theorem yields geometric convergence of the scaled iterates. The identification of the limit with the normalized maximizing potentials concludes the argument. Define the positive linear operators

$$\mathrm{K} : \mathrm{L}^\infty(\mathcal{Y}) \to \mathrm{L}^\infty(\mathcal{X}), \qquad (\mathrm{K}\, b)(x) \triangleq \int_{\mathcal{Y}} e^{-\frac{\mathsf{c}(x,y)}{\alpha}}\, b(y)\, \mathrm{d}Q(y), \tag{110}$$

$$\mathrm{K}^\mathsf{T} : \mathrm{L}^\infty(\mathcal{X}) \to \mathrm{L}^\infty(\mathcal{Y}), \qquad (\mathrm{K}^\mathsf{T}\, a)(y) \triangleq \int_{\mathcal{X}} e^{-\frac{\mathsf{c}(x,y)}{\alpha}}\, a(x)\, \mathrm{d}P(x). \tag{111}$$

Since c is bounded, it holds for all $(x, y) \in \mathcal{X} \times \mathcal{Y}$ that

$$e^{-\frac{\|c\|_\infty}{\alpha}} \;\le\; e^{-\frac{\mathsf{c}(x,y)}{\alpha}} \;\le\; e^{\frac{\|c\|_\infty}{\alpha}}. \tag{112}$$

Hence, for every $b \in \mathrm{L}^\infty_+(\mathcal{Y}) \setminus \{0\}$ and every $x \in \mathcal{X}$,

$$e^{-\frac{\|c\|_\infty}{\alpha}} \int_{\mathcal{Y}} b(y)\, \mathrm{d}Q(y) \;\le\; (\mathrm{K}\, b)(x) \;\le\; e^{\frac{\|c\|_\infty}{\alpha}} \int_{\mathcal{Y}} b(y)\, \mathrm{d}Q(y). \tag{113}$$

Since $\int_{\mathcal{Y}} b(y)\, \mathrm{d}Q(y) > 0$, this yields $\mathrm{K}\, b \in \mathrm{L}^\infty_{++}(\mathcal{X})$. The same argument gives $\mathrm{K}^\mathsf{T}\, a \in \mathrm{L}^\infty_{++}(\mathcal{Y})$ for every $a \in \mathrm{L}^\infty_+(\mathcal{X}) \setminus \{0\}$. If $b_1, b_2 \in \mathrm{L}^\infty_+(\mathcal{Y}) \setminus \{0\}$, then for every $x \in \mathcal{X}$,

$$\frac{(\mathrm{K}\, b_1)(x)}{(\mathrm{K}\, b_2)(x)} \le e^{\frac{2\|c\|_\infty}{\alpha}} \frac{\int_{\mathcal{Y}} b_1(y)\, \mathrm{d}Q(y)}{\int_{\mathcal{Y}} b_2(y)\, \mathrm{d}Q(y)}, \tag{114}$$

$$\frac{(\mathrm{K}\, b_1)(x)}{(\mathrm{K}\, b_2)(x)} \ge e^{-\frac{2\|c\|_\infty}{\alpha}} \frac{\int_{\mathcal{Y}} b_1(y)\, \mathrm{d}Q(y)}{\int_{\mathcal{Y}} b_2(y)\, \mathrm{d}Q(y)}. \tag{115}$$

Taking the essential supremum and the essential infimum over $x \in \mathcal{X}$ yields

$$d_{\mathrm{H}}(\mathrm{K}\, b_1, \mathrm{K}\, b_2) \;\le\; \frac{4\|c\|_\infty}{\alpha}. \tag{116}$$

Hence the projective diameter of K is finite, with

$$\Delta(\mathrm{K}) \le \frac{4\|c\|_\infty}{\alpha}. \tag{117}$$

The same argument yields

$$\Delta(\mathrm{K}^{\mathsf{T}}) \leq \frac{4\|c\|_\infty}{\alpha}. \tag{118}$$

Since $\Delta(\mathrm{K}) < +\infty$ and $\Delta(\mathrm{K}^{\mathsf{T}}) < +\infty$, Theorem 2.9 in Lemmens and Nussbaum [2013] applies to the linear maps

$$\mathrm{K} : \mathrm{L}^\infty(\mathcal{Y}) \to \mathrm{L}^\infty(\mathcal{X}) \qquad \text{and} \qquad \mathrm{K}^{\mathsf{T}} : \mathrm{L}^\infty(\mathcal{X}) \to \mathrm{L}^\infty(\mathcal{Y}), \tag{119}$$

with source and target cones $\mathrm{L}^\infty_+(\mathcal{Y})$ and $\mathrm{L}^\infty_+(\mathcal{X})$. It follows that K is a strict contraction, for Hilbert's metric, from the part $\mathrm{L}^\infty_{++}(\mathcal{Y})$ of $\mathrm{L}^\infty_+(\mathcal{Y})$ into the part $\mathrm{L}^\infty_{++}(\mathcal{X})$ of $\mathrm{L}^\infty_+(\mathcal{X})$, and that K^{T} is a strict contraction, for Hilbert's metric, from $\mathrm{L}^\infty_{++}(\mathcal{X})$ into $\mathrm{L}^\infty_{++}(\mathcal{Y})$, with contraction factors $\tanh(\Delta(\mathrm{K})/4)$ and $\tanh(\Delta(\mathrm{K}^{\mathsf{T}})/4)$, respectively. Hence, for all $b_1, b_2 \in \mathrm{L}^\infty_{++}(\mathcal{Y})$ and all $a_1, a_2 \in \mathrm{L}^\infty_{++}(\mathcal{X})$, it holds that

$$d_{\mathrm{H}}(\mathrm{K}b_1,\mathrm{K}b_2) \leq \tanh\left(\frac{\Delta(\mathrm{K})}{4}\right) d_{\mathrm{H}}(b_1,b_2) \leq \tanh\left(\frac{\|c\|_\infty}{\alpha}\right) d_{\mathrm{H}}(b_1,b_2), \tag{120}$$

$$d_{\mathrm{H}}\Big(\mathrm{K}^{\mathsf{T}}a_1,\mathrm{K}^{\mathsf{T}}a_2\Big) \leq \tanh\left(\frac{\Delta(\mathrm{K}^{\mathsf{T}})}{4}\right) d_{\mathrm{H}}(a_1,a_2) \leq \tanh\left(\frac{\|c\|_\infty}{\alpha}\right) d_{\mathrm{H}}(a_1,a_2). \tag{121}$$

The first scaled update satisfies

$$\Xi(b)(x) = \frac{1}{(\mathrm{K}\, b)(x)}, \tag{122}$$

for every $b \in \mathrm{L}^\infty_{++}(\mathcal{Y})$ and every $x \in \mathcal{X}$. Moreover, relation (30) implies that, for every pair f, g in either $\mathrm{L}^\infty_{++}(\mathcal{X})$ or $\mathrm{L}^\infty_{++}(\mathcal{Y})$,

$$d_{\mathrm{H}}(f^{-1}, g^{-1}) = d_{\mathrm{H}}(f, g). \tag{123}$$

Consequently, for all $b_1, b_2 \in \mathrm{L}^\infty_{++}(\mathcal{Y})$,

$$\begin{aligned} d_{\mathrm{H}}(\Xi(b_1), \Xi(b_2)) &= d_{\mathrm{H}}\big((\mathrm{K}\, b_1)^{-1}, (\mathrm{K}\, b_2)^{-1}\big) && (124)\\ &= d_{\mathrm{H}}(\mathrm{K}\, b_1, \mathrm{K}\, b_2) && (125)\\ &\leq \tanh\left(\frac{\|c\|_\infty}{\alpha}\right) d_{\mathrm{H}}(b_1, b_2). && (126) \end{aligned}$$

Let $a \in \mathrm{L}^\infty_{++}(\mathcal{X})$. The second scaled update can be written as

$$\Gamma(a)(y) = \Big(\mathrm{K}^{\mathsf{T}}\, a(y)\Big)^{-\frac{\beta}{\alpha+\beta}} \left(\int_{\mathcal{Y}} \Big(\mathrm{K}^{\mathsf{T}}\, a(y')\Big)^{\frac{\alpha}{\alpha+\beta}}\, \mathrm{d}Q(y')\right)^{\frac{\beta}{\alpha}}, \tag{127}$$

and $\Gamma(a) \in \mathrm{L}^\infty_{++}(\mathcal{Y})$. The second factor is a positive scalar independent of y. Lemma 2.1(iv) in Lemmens and Nussbaum [2013] therefore gives, for all $a_1, a_2 \in \mathrm{L}^\infty_{++}(\mathcal{X})$,

$$d_{\mathrm{H}}(\Gamma(a_1), \Gamma(a_2)) = d_{\mathrm{H}}\left(\big(\mathrm{K}^{\mathsf{T}}\, a_1\big)^{-\frac{\beta}{\alpha+\beta}}, \big(\mathrm{K}^{\mathsf{T}}\, a_2\big)^{-\frac{\beta}{\alpha+\beta}}\right). \tag{128}$$

The map $f \mapsto f^{\frac{\beta}{\alpha+\beta}}$ is order-preserving and homogeneous of degree $\frac{\beta}{\alpha+\beta}$. Proposition 2.6 in Lemmens and Nussbaum [2013] then implies

$$d_{\mathrm{H}}\Big(f^{\frac{\beta}{\alpha+\beta}}, g^{\frac{\beta}{\alpha+\beta}}\Big) \;\leq\; \frac{\beta}{\alpha+\beta}\, d_{\mathrm{H}}(f, g) \tag{129}$$

for all $f, g \in \mathrm{L}^\infty_{++}(\mathcal{Y})$. Together with (123), this yields

$$\begin{aligned} d_{\mathrm{H}}(\Gamma(a_1), \Gamma(a_2)) &\leq \frac{\beta}{\alpha+\beta}\, d_{\mathrm{H}}\Big(\mathrm{K}^{\mathsf{T}}\, a_1, \mathrm{K}^{\mathsf{T}}\, a_2\Big) && (130)\\ &\leq \frac{\beta}{\alpha+\beta} \tanh\left(\frac{\|c\|_\infty}{\alpha}\right) d_{\mathrm{H}}(a_1, a_2). && (131) \end{aligned}$$

Combining (126) and (131) shows that the one step map $T \triangleq \Gamma \circ \Xi$ satisfies, for all $b_1, b_2 \in \mathrm{L}^\infty_{++}(\mathcal{Y})$,

$$d_{\mathrm{H}}(T(b_1), T(b_2)) \leq \frac{\beta}{\alpha+\beta} \tanh\left(\frac{\|c\|_\infty}{\alpha}\right)^2 d_{\mathrm{H}}(b_1, b_2). \tag{132}$$

It remains to identify a complete metric space left invariant under T. For every $a \in \mathrm{L}^\infty_{++}(\mathcal{X})$ and every $y \in \mathcal{Y}$,

$$\Gamma(a)(y)^{-\alpha/\beta} = \frac{\left(\mathrm{K}^\mathsf{T}\, a(y)\right)^{\frac{\alpha}{\alpha+\beta}}}{\int_{\mathcal{Y}} \left(\mathrm{K}^\mathsf{T}\, a(y')\right)^{\frac{\alpha}{\alpha+\beta}}\, \mathrm{d}Q(y')}. \tag{133}$$

Integration with respect to Q yields

$$\int_{\mathcal{Y}} \Gamma(a)(y)^{-\alpha/\beta}\, \mathrm{d}Q(y) = 1. \tag{134}$$

Hence every iterate $b^{(l)}$ belongs to

$$\Sigma \triangleq \left\{ b \in \mathrm{L}^\infty_{++}(\mathcal{Y}) : \int_{\mathcal{Y}} b(y)^{-\alpha/\beta}\, \mathrm{d}Q(y) = 1 \right\}. \tag{135}$$

Consider

$$q(b) \triangleq \left(\int_{\mathcal{Y}} b(y)^{-\alpha/\beta}\, \mathrm{d}Q(y) \right)^{-\beta/\alpha}. \tag{136}$$

This map is homogeneous of degree 1. It is also continuous on $\mathrm{L}^\infty_{++}(\mathcal{Y})$ for the L^∞ norm. Indeed, let b be in $\mathrm{L}^\infty_{++}(\mathcal{Y})$ and $(b_n)_{n\geq 1}$ be a sequence in $\mathrm{L}^\infty_{++}(\mathcal{Y})$ such that $b_n \to b$ in $\mathrm{L}^\infty(\mathcal{Y})$, and set

$$m_b \triangleq \operatorname*{ess\,inf}_{Q} b > 0. \tag{137}$$

There exists $n_0 \in \mathbb{N}$ such that $b_n(y) \geq m_b/2$ for Q-almost every $y \in \mathcal{Y}$ and every $n \geq n_0$. The map $t \mapsto t^{-\alpha/\beta}$ is Lipschitz on $[m_b/2, +\infty)$, hence

$$\left\| b_n^{-\alpha/\beta} - b^{-\alpha/\beta} \right\|_\infty \xrightarrow[n\to+\infty]{} 0. \tag{138}$$

Therefore

$$\int_{\mathcal{Y}} b_n(y)^{-\alpha/\beta}\, \mathrm{d}Q(y) \xrightarrow[n\to+\infty]{} \int_{\mathcal{Y}} b(y)^{-\alpha/\beta}\, \mathrm{d}Q(y), \tag{139}$$

and then $q(b_n) \to q(b)$. The cone $\mathrm{L}^\infty_{+}(\mathcal{Y})$ is closed and normal in the Banach space $\mathrm{L}^\infty(\mathcal{Y})$, and $\mathrm{L}^\infty_{++}(\mathcal{Y})$ is the part containing the constant function 1. Lemma 2.1 in Nussbaum [2007], applied with the constant function 1, yields that (Σ, d_{H}) is a complete metric space. Since $b^{(0)} \equiv 1$ belongs to Σ and $T(\Sigma) \subset \Sigma$ follows from (134), the Banach fixed point theorem implies the existence of a unique fixed point $\hat{b} \in \Sigma$ and gives, for all $l \geq 0$,

$$d_{\mathrm{H}}\left(b^{(l)}, \hat{b}\right) \leq \left(\frac{\beta}{\alpha+\beta} \tanh\left(\frac{\|c\|_\infty}{\alpha} \right)^2 \right)^l d_{\mathrm{H}}\left(b^{(0)}, \hat{b}\right), \tag{140}$$

which is exactly (33). Define

$$\hat{a} \triangleq \Xi(\hat{b}). \tag{141}$$

Then, for every $l \geq 1$,

$$\begin{aligned} d_{\mathrm{H}}\left(a^{(l)}, \hat{a}\right) &= d_{\mathrm{H}}\left(\Xi(b^{(l-1)}), \Xi(\hat{b})\right) && (142)\\ &\leq \tanh\left(\frac{\|c\|_\infty}{\alpha}\right) d_{\mathrm{H}}\left(b^{(l-1)}, \hat{b}\right) && (143)\\ &\leq \tanh\left(\frac{\|c\|_\infty}{\alpha}\right) \rho^{l-1}_{(\mathsf{c},\alpha,\beta)}\, d_{\mathrm{H}}\left(b^{(0)}, \hat{b}\right), && (144) \end{aligned}$$

which is exactly (34).

Finally, define

$$\hat{u} \triangleq \alpha \log(\hat{a}) \in \mathrm{L}^\infty(\mathcal{X}), \qquad \hat{v} \triangleq \alpha \log(\hat{b}) \in \mathrm{L}^\infty(\mathcal{Y}). \tag{145}$$

The fixed point relation $T(\hat{b}) = \hat{b}$ and the definition $\hat{a} = \Xi(\hat{b})$ are equivalent to

$$\hat{u} = \Phi(\hat{v}), \qquad \hat{v} = \Psi(\hat{u}), \tag{146}$$

where Φ and Ψ are defined in (16) and (18). Corollary 3.1 and Corollary 3.3 then imply that $\hat{u}$ and $\hat{v}$ are the unique maximizers for problem (10) that satisfy (15) and (17) simultaneously. Therefore, $\hat{a}$ and $\hat{b}$ are the associated unique scaled maximizing potentials. ■

A.12 Proof of Corollary 3.4

Proof: The proof is split into three parts; the first one deals with (36), the second one with (37), and the third one with (40). Let $(u^{(l)}, v^{(l)})$ be the unscaled iterates defined by Algorithm 1, $(a^{(l)}, b^{(l)})$ be the scaled iterates defined by Algorithm 2, and $(\hat{u}, \hat{v})$ be the unique pair of maximizers satisfying (15) and (17) simultaneously. Recall that the two parametrizations are linked through

$$u^{(l)} = \alpha \log\big(a^{(l)}\big), \qquad v^{(l)} = \alpha \log\big(b^{(l)}\big). \tag{147}$$

Hence, it holds that

$$\|v^{(l)} - \hat{v}\|_\infty = \alpha \left\| \log\left(\frac{b^{(l)}}{\hat{b}}\right) \right\|_\infty . \tag{148}$$

Define the pointwise ratio

$$r^{(l)}(y) \triangleq \frac{b^{(l)}(y)}{\hat{b}(y)}, \qquad y \in \mathcal{Y}, \tag{149}$$

and its bounds

$$m^{(l)} \triangleq \inf_{y \in \mathcal{Y}} r^{(l)}(y), \qquad M^{(l)} \triangleq \sup_{y \in \mathcal{Y}} r^{(l)}(y). \tag{150}$$

By definition of Hilbert's projective metric on $\mathrm{L}^\infty_{++}(\mathcal{Y})$ (Definition 3.1), it holds that

$$d_{\mathrm{H}}\big(b^{(l)}, \hat{b}\big) = \log\left(\frac{M^{(l)}}{m^{(l)}}\right). \tag{151}$$

It remains to show that

$$m^{(l)} \le 1 \le M^{(l)}. \tag{152}$$

Since the iterates $b^{(l)}$ are in Σ for all l and $\hat{b} \in \Sigma$, where Σ is in (135), the same gauge constraint as in (134) holds

$$\int_{\mathcal{Y}} \big(b^{(l)}(y)\big)^{-\alpha/\beta} \, \mathrm{d}Q(y) = 1, \qquad \text{and} \int_{\mathcal{Y}} \big(\hat{b}(y)\big)^{-\alpha/\beta} \, \mathrm{d}Q(y) = 1. \tag{153}$$

The map $t \mapsto t^{-\alpha/\beta}$ is strictly decreasing on $(0, +\infty)$. Assume by contradiction that $M^{(l)} < 1$. Then $r^{(l)}(y) < 1$ for Q-a.e. y, i.e., $b^{(l)}(y) < \hat{b}(y)$ for Q-a.e. y, and the strict monotonicity of the integral implies

$$\int_{\mathcal{Y}} \big(b^{(l)}(y)\big)^{-\alpha/\beta} \, \mathrm{d}Q(y) > \int_{\mathcal{Y}} \big(\hat{b}(y)\big)^{-\alpha/\beta} \, \mathrm{d}Q(y), \tag{154}$$

which contradicts (153). Therefore $M^{(l)} \ge 1$. The case $m^{(l)} > 1$ is ruled out similarly. Thus, the inequalities in (152) are established. Combining (152) with $\|\log r^{(l)}\|_\infty = \max\{-\log m^{(l)}, \log M^{(l)}\}$ gives

$$\left\| \log\left(\frac{b^{(l)}}{\hat{b}}\right) \right\|_\infty = \max\{-\log m^{(l)}, \log M^{(l)}\} \tag{155}$$

$$\le \log\left(\frac{M^{(l)}}{m^{(l)}}\right) \tag{156}$$

$$= d_{\mathrm{H}}\big(b^{(l)}, \hat{b}\big), \tag{157}$$

where the last equality follows from (151). Plugging (157) into (148) yields

$$\|v^{(l)} - \hat{v}\|_\infty \le \alpha \, d_{\mathrm{H}}\big(b^{(l)}, \hat{b}\big). \tag{158}$$

Combining the previous inequality with (33) concludes the proof of (36). Regarding the proof of (37), recall from Algorithm 1 that

$$u^{(l+1)} = \Phi\big(v^{(l)}\big), \qquad \text{and } \hat{u} = \Phi(\hat{v}), \tag{159}$$

where Φ is in (16). Let v, w be in $\mathrm{L}^\infty(\mathcal{Y})$ and set $\delta \triangleq \|v - w\|_\infty$. Then, for Q-a.e. y in $\mathcal{Y}$, it holds that

$$-\delta \leq v(y) - w(y) \leq \delta \quad \Longrightarrow \quad e^{-\delta/\alpha} \leq \exp\left(\frac{v(y) - w(y)}{\alpha}\right) \leq e^{\delta/\alpha}. \tag{160}$$

Multiplying by $\exp\left(\frac{w(y)-\mathsf{c}(x,y)}{\alpha}\right)$ and integrating with respect to Q yields, for every $x \in \mathcal{X}$,

$$e^{-\delta/\alpha} \int_{\mathcal{Y}} \exp\left(\frac{w(y)-\mathsf{c}(x,y)}{\alpha}\right) \mathrm{d}Q(y) \leq \int_{\mathcal{Y}} \exp\left(\frac{v(y)-\mathsf{c}(x,y)}{\alpha}\right) \mathrm{d}Q(y) \tag{161}$$

$$\leq e^{\delta/\alpha} \int_{\mathcal{Y}} \exp\left(\frac{w(y)-\mathsf{c}(x,y)}{\alpha}\right) \mathrm{d}Q(y). \tag{162}$$

Applying $-\alpha$ times the logarithm to the previous inequalities yields, for every $x \in \mathcal{X}$,

$$\Phi(w)(x) - \delta \leq \Phi(v)(x) \leq \Phi(w)(x) + \delta, \tag{163}$$

hence

$$\|\Phi(v) - \Phi(w)\|_\infty \leq \|v - w\|_\infty. \tag{164}$$

Combining (159) with (164) yields

$$\|u^{(l+1)} - \hat{u}\|_\infty = \|\Phi(v^{(l)}) - \Phi(\hat{v})\|_\infty \leq \|v^{(l)} - \hat{v}\|_\infty. \tag{165}$$

Together with (36), this implies

$$\|u^{(l+1)} - \hat{u}\|_\infty \leq \alpha \, d_{\mathrm{H}}\big(b^{(l)}, \hat{b}\big). \tag{166}$$

Finally, applying (33) concludes the proof of (37). ■

A.13 Proof of Corollary 3.5

Proof: From Corollary 3.1, each alternating maximization step does not decrease the value of D, hence $\mathsf{D}(u^{(l)}, v^{(l)}) \leq \mathsf{D}(u^{(l+1)}, v^{(l+1)})$ for all $l \geq 0$. Strong duality, established in Theorem 3.2, provides the left-hand inequality in (38). The right-hand inequality rests on the following observation: for any measurable bounded functions f and g on a probability space (Ω, μ), it holds that

$$\left|\log \int_\Omega e^{f(t)} \, \mathrm{d}\mu(t) \; - \; \log \int_\Omega e^{g(t)} \, \mathrm{d}\mu(t)\right| \; \leq \; \|f - g\|_\infty. \tag{167}$$

Indeed, $\int e^{f(t)} \, \mathrm{d}\mu(t) \leq e^{\|f-g\|_\infty} \int e^{g(t)} \, \mathrm{d}\mu(t)$, from which it can be deduced that

$$\log \int_\Omega e^{f(t)} \, \mathrm{d}\mu(t) - \log \int_\Omega e^{g(t)} \, \mathrm{d}\mu(t) \leq \|f - g\|_\infty. \tag{168}$$

The reverse inequality is obtained symmetrically. From definition (10), the gap decomposes as

$$\begin{aligned}
\mathsf{D}(\hat{u},\hat{v}) - \mathsf{D}(u^{(l)},v^{(l)}) = & \int_{\mathcal{X}} \big(\hat{u} - u^{(l)}\big) \, \mathrm{d}P \\
& - \beta \left[\log \int_{\mathcal{Y}} e^{-\hat{v}(y)/\beta} \, \mathrm{d}Q(y) \; - \; \log \int_{\mathcal{Y}} e^{-v^{(l)}(y)/\beta} \, \mathrm{d}Q(y)\right] \\
& - \alpha \Bigg[\log \int_{\mathcal{X}\times\mathcal{Y}} e^{(\hat{u}(x)+\hat{v}(y)-\mathsf{c}(x,y))/\alpha} \mathrm{d}PQ(x,y) \\
& \qquad - \log \int_{\mathcal{X}\times\mathcal{Y}} e^{(u^{(l)}(x)+v^{(l)}(y)-\mathsf{c}(x,y))/\alpha} \mathrm{d}PQ(x,y)\Bigg].
\end{aligned} \tag{169}$$

Applying (167) to the second and third terms of (169), and using the fact that P is a probability measure for the first term, yields

$$\mathsf{D}(\hat{u}, \hat{v}) - \mathsf{D}(u^{(l)}, v^{(l)}) \; \leq \; 2\left\|\hat{u} - u^{(l)}\right\|_\infty \; + \; 2\left\|\hat{v} - v^{(l)}\right\|_\infty. \tag{170}$$

Applying (37) and (36) to the right-hand side of (170) concludes the proof. ■

A.14 Proof of Corollary 3.6

Proof: The proof establishes successively that $\pi^{(l)}$ is in $\Pi(P,\star)$, that the relevant exponents are uniformly bounded, and that this bound combined with the linear convergence of the potentials yields (40). Since $u^{(l+1)} = \Phi(v^{(l)})$ in Algorithm 1, the definition (16) of Φ gives, for every $x \in \mathcal{X}$,

$$\begin{aligned}\int_{\mathcal{Y}} \frac{\mathrm{d}\pi^{(l)}}{\mathrm{d}PQ}(x,y)\,\mathrm{d}Q(y) &= e^{u^{(l+1)}(x)/\alpha} \int_{\mathcal{Y}} e^{(v^{(l)}(y)-\mathsf{c}(x,y))/\alpha}\,\mathrm{d}Q(y) \\ &= e^{u^{(l+1)}(x)/\alpha} \cdot e^{-u^{(l+1)}(x)/\alpha} \;=\; 1. \end{aligned} \tag{171}$$

Fubini's theorem [Rudin, 1987, Theorem 8.8] then implies that $\pi^{(l)}$ is a probability measure with first marginal $(\pi^{(l)})^{\mathcal{X}}_{\#} = P$, hence $\pi^{(l)} \in \Pi(P,\star)$. From (22), it holds that $\|v^{(l)}\|_\infty \le M\|\mathsf{c}\|_\infty$ for all $l \ge 1$, and from (21), $\|u^{(l+1)}\|_\infty \le \|v^{(l)}\|_\infty + \|\mathsf{c}\|_\infty \le (M+1)\|\mathsf{c}\|_\infty$ for all $l \ge 1$. The same estimates hold for $\hat{u}$ and $\hat{v}$, since $(\hat{u},\hat{v})$ satisfies (15) and (17). Hence, for all $(x,y) \in \mathcal{X}\times\mathcal{Y}$ and all $l \ge 0$, the two arguments

$$\frac{u^{(l+1)}(x) + v^{(l)}(y) - \mathsf{c}(x,y)}{\alpha} \quad \text{and} \quad \frac{\hat{u}(x) + \hat{v}(y) - \mathsf{c}(x,y)}{\alpha} \tag{172}$$

are both bounded in absolute value by $(2M+2)\|\mathsf{c}\|_\infty/\alpha$. For real numbers a and b with $|a|, |b| \le R$, the mean value theorem gives $|e^a - e^b| \le e^R|a-b|$. Applying this pointwise with $R = (2M+2)\|\mathsf{c}\|_\infty/\alpha$ and using (172), for all $(x,y) \in \mathcal{X}\times\mathcal{Y}$ it holds that

$$\begin{aligned}&\left|\frac{\mathrm{d}\pi^{(l)}}{\mathrm{d}PQ}(x,y) - \frac{\mathrm{d}\hat{\pi}}{\mathrm{d}PQ}(x,y)\right| \\ &\le \frac{e^{(2M+2)\|\mathsf{c}\|_\infty/\alpha}}{\alpha}\left(\left|u^{(l+1)}(x) - \hat{u}(x)\right| + \left|v^{(l)}(y) - \hat{v}(y)\right|\right). \end{aligned} \tag{173}$$

Applying (37) and (36) to the previous expression yields

$$\left\|\frac{\mathrm{d}\pi^{(l)}}{\mathrm{d}PQ} - \frac{\mathrm{d}\hat{\pi}}{\mathrm{d}PQ}\right\|_\infty \le 2\,e^{(2M+2)\|\mathsf{c}\|_\infty/\alpha}\,\rho^{l}_{(\mathsf{c},\alpha,\beta)}\,d_{\mathrm{H}}\left(b^{(0)},\hat{b}\right), \tag{174}$$

which gives (40). The total variation bound then follows from

$$\left\|\pi^{(l)} - \hat{\pi}\right\|_{TV} \;\triangleq\; \frac{1}{2}\int_{\mathcal{X}\times\mathcal{Y}}\left|\frac{\mathrm{d}\pi^{(l)}}{\mathrm{d}PQ} - \frac{\mathrm{d}\hat{\pi}}{\mathrm{d}PQ}\right|\mathrm{d}PQ \;\le\; \frac{1}{2}\left\|\frac{\mathrm{d}\pi^{(l)}}{\mathrm{d}PQ} - \frac{\mathrm{d}\hat{\pi}}{\mathrm{d}PQ}\right\|_\infty, \tag{175}$$

since PQ is a probability measure. ■

NeurIPS Paper Checklist

The checklist is designed to encourage best practices for responsible machine learning research, addressing issues of reproducibility, transparency, research ethics, and societal impact. Do not remove the checklist: **The papers not including the checklist will be desk rejected.** The checklist should follow the references and follow the (optional) supplemental material. The checklist does NOT count towards the page limit.

Please read the checklist guidelines carefully for information on how to answer these questions. For each question in the checklist:

- You should answer [Yes], [No], or [N/A].
- [N/A] means either that the question is Not Applicable for that particular paper or the relevant information is Not Available.
- Please provide a short (1–2 sentence) justification right after your answer (even for [N/A]).

The checklist answers are an integral part of your paper submission. They are visible to the reviewers, area chairs, senior area chairs, and ethics reviewers. You will also be asked to include it (after eventual revisions) with the final version of your paper, and its final version will be published with the paper.

The reviewers of your paper will be asked to use the checklist as one of the factors in their evaluation. While [Yes] is generally preferable to [No], it is perfectly acceptable to answer [No] provided a proper justification is given (e.g., error bars are not reported because it would be too computationally expensive" or "we were unable to find the license for the dataset we used"). In general, answering [No] or [N/A] is not grounds for rejection. While the questions are phrased in a binary way, we acknowledge that the true answer is often more nuanced, so please just use your best judgment and write a justification to elaborate. All supporting evidence can appear either in the main paper or the supplemental material, provided in appendix. If you answer [Yes] to a question, in the justification please point to the section(s) where related material for the question can be found.

IMPORTANT, please:

- **Delete this instruction block, but keep the section heading "NeurIPS Paper Checklist"**,
- **Keep the checklist subsection headings, questions/answers and guidelines below.**
- **Do not modify the questions and only use the provided macros for your answers**.

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: **[TODO]**

   Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate "Limitations" section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not include experiments.

- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
  (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).

- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
   - The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
   - The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
   - The assumptions made should be given (e.g., Normally distributed errors).
   - It should be clear whether the error bar is the standard deviation or the standard error of the mean.
   - It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
   - For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
   - If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.

- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper poses no such risks.

- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that the paper does not use existing assets.
    - The authors should cite the original paper that produced the code package or dataset.
    - The authors should state which version of the asset is used and, if possible, include a URL.
    - The name of the license (e.g., CC-BY 4.0) should be included for each asset.
    - For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
    - If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
    - For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
    - If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
    - The paper should discuss whether and how consent was obtained from people whose asset is used.
    - At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: **[TODO]**

    Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.